Predicting Employee Turnover: A Statistical Analysis of Workforce Attrition Factors

Brenda Murillo
Dept. of Computer and Electrical Engineering and Computer Science
California State University, Bakersfield
Bakersfield, CA, USA
bmurillo3@csub.edu

***Abstract*—Employee turnover represents one of the most persistent and costly challenges facing organizations today, yet most retention strategies rely on intuition rather than data. This paper presents a statistical analysis of workforce attrition using an HR dataset of 14,999 employee records drawn from Kaggle. We apply descriptive statistics, relational database querying (SQLite), correlation analysis, and visual analysis to identify the factors most strongly associated with employee departure. Our findings reveal an overall turnover rate of 23.81%, with low-salary employees leaving at nearly 30%—more than four times the rate of high-salary employees. Job satisfaction emerges as the single strongest predictor ($r = -0.388$); employees who left averaged a satisfaction score of 0.44 compared to 0.67 for those who stayed. We further identify 615 current employees who match a high-risk churn profile. These findings provide HR departments with actionable, data-driven criteria for early intervention and targeted retention strategies.**

***Index Terms*—employee turnover, workforce attrition, HR analytics, churn prediction, descriptive statistics, SQLite, data analysis**

I. INTRODUCTION

The cost of replacing an employee is widely estimated at between 50% and 200% of that employee's annual salary, accounting for recruiting, onboarding, lost productivity, and institutional knowledge transfer [2]. Yet despite this substantial burden, many organizations respond to turnover reactively—conducting exit interviews only after employees have already resigned, and implementing changes too late to retain the individuals who prompted them.

The central challenge is one of prediction: given observable workplace factors, can an organization identify which employees are most likely to leave before they submit their resignation? This problem is well-suited to data-driven analysis. Modern HR systems routinely capture variables such as satisfaction surveys, performance evaluations, compensation tiers, workload metrics, and promotion histories—precisely the kinds of signals that may indicate attrition risk.

This paper addresses that challenge by applying statistical analysis to an HR dataset containing records for 14,999 employees. We pursue four core research questions: (1) How strongly does salary level predict turnover? (2) What is the relationship between job satisfaction and departure? (3) Does workload influence turnover? (4) Can we identify current employees at high churn risk using observable characteristics?

Our approach is grounded in accessible statistical methods—descriptive statistics, group comparisons, correlation analysis, and SQL querying against a normalized relational database—rather than black-box machine learning models, making our findings interpretable and actionable for HR practitioners.

II. RELATED WORK

Employee attrition prediction has been an active area of research in both human resources management and data science. Early work by Mobley [1] established a foundational behavioral model of turnover, identifying job dissatisfaction as the primary precursor to resignation. Griffeth, Hom, and Gaertner [2] confirmed through meta-analysis that satisfaction, pay, and workload are the most reliable predictors of departure across industries.

More recent computational approaches have applied supervised machine learning to HR datasets. Punnoose and Ajit [5] applied random forests and logistic regression to the same Kaggle HR dataset, achieving accuracies above 97%. Alduayj and Rajpoot [7] compared decision trees, SVMs, and neural networks, finding tree-based models consistently outperformed others on HR tabular data. Zhao et al. [6] demonstrated ensemble methods improve performance but at significant cost to interpretability [8].

Our work is positioned in the tradition of transparent statistical approaches that make decision criteria visible and reproducible by any analyst, prioritizing interpretability over predictive accuracy.

III. DATASET

A. Source and Scope

The dataset is the HR Employee Retention dataset publicly available on Kaggle [9], compiled to simulate realistic human resources records. It contains 14,999 employee records, each representing a unique individual. The outcome variable (left) indicates whether the employee departed (1) or remained (0).

B. Variables

The dataset includes ten variables spanning four conceptual domains, shown in Table I.

TABLE I. DATASET VARIABLES

Domain	Variable	Type	Description
Satisfaction	satisfaction_level	Numeric [0,1]	Self-reported satisfaction score
Performance	last_evaluation	Numeric [0,1]	Most recent performance review
Workload	number_project	Integer	Number of projects assigned
Workload	average_monthly_hours	Integer	Avg. hours worked per month
Tenure	time_spend_company	Integer	Years employed at company
Safety	work_accident	Boolean	Had a work accident
Career	promotion_last_5years	Boolean	Promoted in last 5 years
Compensation	salary	Categorical	Low / medium / high
Outcome	left	Boolean	Left the company

C. Data Quality

A missing value audit revealed no null entries in any column. Outlier detection using the IQR method identified anomalous values in average_monthly_hours (above 310 hrs/month), retained as plausible extreme-workload cases. Schema constraints enforced CHECK ranges on satisfaction and evaluation scores ([0,1]), non-negative hours and tenure, and foreign key consistency across all four tables. A row-count reconciliation confirmed all 14,999 records are present after normalization.

D. Database Schema

The flat CSV was normalized into a four-table relational schema in SQLite [10]: (1) salary_levels—lookup table mapping salary tier labels to IDs; (2) employees—core identity table (emp_id, salary_id, tenure); (3) performance_records—satisfaction, evaluation, project count, monthly hours; (4) employment_events—work accidents, promotion history, and turnover outcome. This design eliminates redundancy, enforces referential integrity via foreign keys, and allows the turnover outcome to be joined against any combination of attributes in a single SQL query.

IV. METHODOLOGY

A. Descriptive Statistics

For all numeric variables, we computed mean, median, mode, standard deviation, variance, range, Q1, Q3, and the interquartile range, calculated separately for employees who left and those who stayed.

B. Group Comparison

Turnover rates were computed for each level of categorical variables (salary tier, promotion status, work accident). For numeric variables, employees were segmented into bins and turnover rates computed within each bin to identify non-linear relationships.

C. Correlation Analysis

Pearson correlation coefficients were computed between each numeric feature and the binary left outcome, producing a ranked list of predictors by association strength.

D. SQL-Based Analysis

All group-level queries were executed against the normalized SQLite database, including overall turnover rate, turnover by salary tier (three-table join), average satisfaction and monthly hours by turnover status, and high-risk employee identification using compound WHERE filters.

E. Risk Profiling

A high-risk profile was defined using three simultaneous criteria: satisfaction score below 0.4, salary tier of "low", and no promotion in the last five years, constrained to employees still at the company. This threshold was set based on the distributional boundary at which turnover rates were observed to increase sharply.

V. RESULTS

A. Overall Turnover Rate

Of 14,999 employees, 3,571 left the organization, yielding an overall turnover rate of 23.81%. This substantially exceeds the commonly cited U.S. industry average of approximately 15% [4].

B. Turnover by Salary Tier

Salary tier is one of the strongest single-variable predictors of turnover. Table II shows that low-salary employees leave at more than four times the rate of high-salary employees.

TABLE II. TURNOVER RATE BY SALARY TIER

Salary Tier	Total Employees	Left	Turnover Rate
Low	7,316	2,172	29.69%
Medium	6,446	1,317	20.43%
High	1,237	82	6.63%

C. Job Satisfaction and Turnover

Job satisfaction shows the most pronounced difference between retained and departed employees (Table III). It produces the strongest Pearson correlation with turnover ($r = -0.388$). Satisfaction differs by 0.227 points between groups—a 34% relative gap. By contrast, performance evaluation scores are nearly identical (0.715 vs. 0.718), confirming that turnover is a satisfaction problem, not a performance problem.

TABLE III. AVERAGE FEATURE VALUES BY TURNOVER STATUS

Feature	Stayed	Left
Satisfaction level	0.667	0.440
Last evaluation score	0.715	0.718
Number of projects	3.79	3.86
Avg. monthly hours	199.1	207.4
Years at company	3.38	3.88

D. Feature Correlation Analysis

Table IV presents the Pearson correlation of each feature with the left outcome, ranked by absolute magnitude. Satisfaction is the strongest predictor by a wide margin (r = −0.388). last_evaluation shows virtually no correlation (r = +0.007), confirming that performance evaluation does not distinguish leavers from stayers.

TABLE IV. FEATURE CORRELATION WITH TURNOVER

Feature	Pearson r
satisfaction_level	−0.388
work_accident	−0.155
time_spend_company	+0.145
average_monthly_hours	+0.071
promotion_last_5years	−0.062
number_project	+0.024
last_evaluation	+0.007

E. Workload and Turnover

Employees who left worked an average of 207.4 hours per month compared to 199.1 for those who stayed—a difference of 8.3 hours, or roughly two extra workdays per month. The correlation of average_monthly_hours with turnover (r = +0.071) suggests overwork is a contributing factor, though it acts as a compounding stressor alongside low satisfaction and low pay rather than an independent primary driver.

F. High-Risk Employee Profiling

Applying the compound risk profile (satisfaction < 0.4, salary = low, no promotion in 5 years, still employed), the analysis identified 615 current employees who match all three criteria. Table V shows the ten highest-risk individuals by lowest satisfaction score.

TABLE V. SAMPLE HIGH-RISK EMPLOYEES (TOP 10)

Emp. ID	Satisfaction	Hrs/Mo	Tenure (yrs)
2087	0.12	244	5
3129	0.12	257	6
4026	0.12	241	2
4684	0.12	276	4
5336	0.12	166	3
7772	0.12	161	4
8745	0.12	287	4
9283	0.12	200	3
11069	0.12	191	5
13280	0.12	191	5

VI. DISCUSSION

A. *Compensation is the Dominant Structural Factor*

The four-to-one ratio in turnover rates between low and high salary tiers is the single most actionable finding in this study. It suggests that compensation structure creates a systemic attrition risk that cannot be resolved through engagement programs or managerial coaching alone. Organizations with large proportions of low-salary employees should treat compensation review as a primary retention strategy [2].

B. *Satisfaction Predicts Turnover Independently of Performance*

The near-identical evaluation scores between leavers and stayers (0.718 vs. 0.715), combined with the near-zero correlation of last_evaluation ($r = +0.007$), challenges the common assumption that turnover is driven by underperformers. Departed employees were equally well-evaluated as retained ones. This replicates Mobley's foundational finding [1] and underscores that retention efforts should be framed as a satisfaction and engagement problem, not a performance management problem.

C. *Workload Acts as a Compounding Risk Factor*

The highest-risk employees combine low satisfaction with above-average hours, suggesting overwork amplifies the effect of low satisfaction rather than driving attrition independently. HR interventions should target the intersection of these factors, not either factor in isolation [3].

D. *Early Intervention Opportunity*

The identification of 615 high-risk current employees is the most practically valuable output of this analysis. Unlike post-departure analyses, this profiling approach is actionable before attrition occurs. HR departments can operationalize it as a recurring SQL query to generate a prioritized list for manager outreach, salary review, or promotion consideration.

E. *Limitations*

This study has several limitations. First, the dataset is simulated [9] and does not represent a specific organization, limiting generalizability. Second, the analysis is descriptive rather than causal: we identify associations, not causal effects. Third, the dataset lacks department or role information, which likely moderates many relationships observed. Fourth, class imbalance (~24% leavers) should be addressed before extending this framework to predictive modeling.

VII. CONCLUSION

This paper analyzed an HR dataset of 14,999 employees to identify the factors most strongly associated with workforce attrition. Salary tier and job satisfaction are the two dominant predictors of turnover—with satisfaction producing the strongest correlation ($r = -0.388$)—while performance evaluation is essentially uncorrelated with departure ($r = +0.007$). Critically, departed employees were equally well-evaluated as retained employees, confirming that turnover is a satisfaction problem, not a performance problem.

By normalizing a flat HR CSV into a constraint-validated SQLite schema and operationalizing findings as reusable SQL queries, we demonstrate that organizations can generate actionable, prioritized intervention lists from existing HR data without requiring machine learning infrastructure. The 615 high-risk employees identified here represent an immediately addressable retention opportunity.

Future work should apply this framework to real organizational data with department-level granularity, incorporate time-series analysis to model attrition risk over tenure, and explore causal inference methods to move beyond association toward actionable causal claims about the effect of salary increases or promotion on retention probability.

ACKNOWLEDGMENT

The author thanks the faculty of the Department of Computer and Electrical Engineering and Computer Science at California State University, Bakersfield for guidance on this project.

REFERENCES

[1] W. H. Mobley, "Intermediate linkages in the relationship between job satisfaction and employee turnover," J. Appl. Psychol., vol. 62, no. 2, pp. 237–240, 1977.

[2] R. W. Griffeth, P. W. Hom, and S. Gaertner, "A meta-analysis of antecedents and correlates of employee turnover," J. Manag., vol. 26, no. 3, pp. 463–488, 2000.

[3] P. W. Hom, T. W. Lee, J. D. Shaw, and J. P. Hausknecht, "One hundred years of employee turnover theory and research," J. Appl. Psychol., vol. 102, no. 3, pp. 530–545, 2017.

[4] Bureau of Labor Statistics, "Job Openings and Labor Turnover Summary," U.S. Department of Labor, 2023.

[5] R. Punnoose and P. Ajit, "Prediction of employee turnover in organizations using machine learning algorithms," Int. J. Adv. Res. Artif. Intell., vol. 5, no. 9, pp. 22–26, 2016.

[6] Y. Zhao, M. K. Hryniewicki, F. Cheng, B. Fu, and X. Zhu, "Employee turnover prediction with machine learning: A reliable approach," in Proc. IntelliSys, 2018, pp. 737–758.

[7] S. S. Alduayj and K. Rajpoot, "Predicting employee attrition using machine learning," in Proc. IIT, 2018, pp. 93–98.

[8] F. Doshi-Velez and B. Kim, "Towards a rigorous science of interpretable machine learning," arXiv:1702.08608, 2017.

[9] HR Analytics / Employee Retention Dataset, Kaggle. [Online]. Available: https://www.kaggle.com/

[10] E. F. Codd, "A relational model of data for large shared data banks," Commun. ACM, vol. 13, no. 6, pp. 377–387, 1970.